GAME THEORY & NASH EQUILIBRIA — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Game Theory & Nash Equilibria

Question 1

10.0 points

Analyze the extensive form game tree in the diagram below for a sequential two-player game where Player 1 can choose between actions A and B, and Player 2 can then choose between X and Y after observing Player 1's action. Determine the subgame perfect Nash equilibrium by performing backward induction on this game tree.

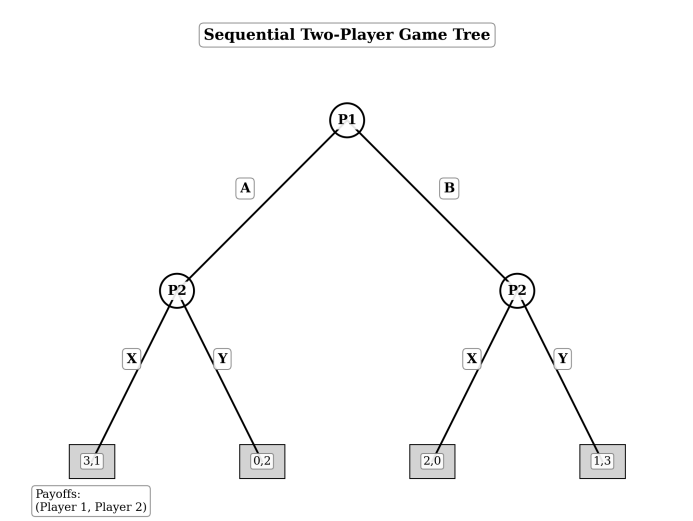

Fig. 1. Extensive form game tree for a sequential two-player game where Player 1 chooses between actions A and B, and Player 2 chooses between X and Y after observing Player 1's action. The diagram illustrates the decision nodes and possible outcomes for performing backward induction.

Question 2

10.0 points

Consider a payoff matrix for a Prisoner's Dilemma: | | Cooperate | Defect | |------- |----------|--------| | C | (3,3) | (0,5) | | D | (5,0) | (1,1) | Identify all Nash equilibria in this game and explain why they are Nash equilibria.

Question 3

10.0 points

Evaluate a mixed strategy Nash equilibrium in a coordination game given by the payoff table in the diagram below, where both players choose high or low effort: | | High | Low | |----------|-------|-------| | High | (2,2) | (0,3) | | Low | (3,0) | (1,1) | Use the mixed strategy method to find the equilibria, and determine the probabilities with which each player should randomize their strategies.

Question 4

10.0 points

Perform backward induction on the game tree shown in the diagram below for a three-stage sequential game with perfect information. Each player has one possible optimal move at each decision node. Determine the optimal strategy for each player, and explain how backward induction aids in determining this strategy.

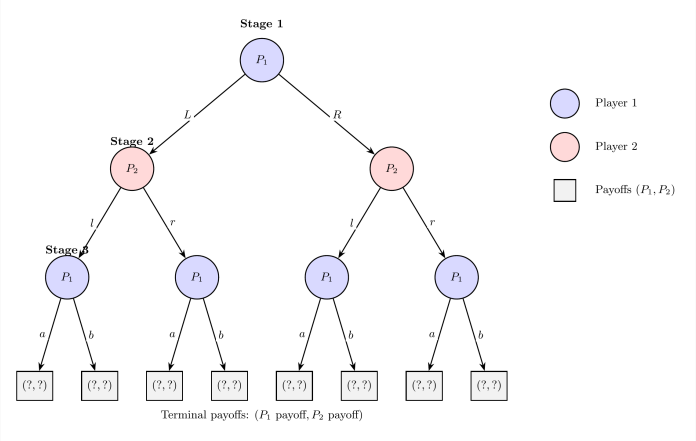

Fig. 4. A three-stage sequential game tree with decision nodes for Player 1 and Player 2, illustrating possible moves at each stage. IMPORTANT: Values or sequences in the diagram don't match the question. The question text and diagram description refer to. Regenerate using EXACTLY the values and sequences specified. IMPORTANT correction needed: The diagram contains annotations on the right. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 5 10.0 points

Given the Pareto frontier illustrated in the diagram below, identify the Pareto efficient points and describe the trade-offs at inefficient points. Explain how any shift in the curve can alter efficiency in allocation.

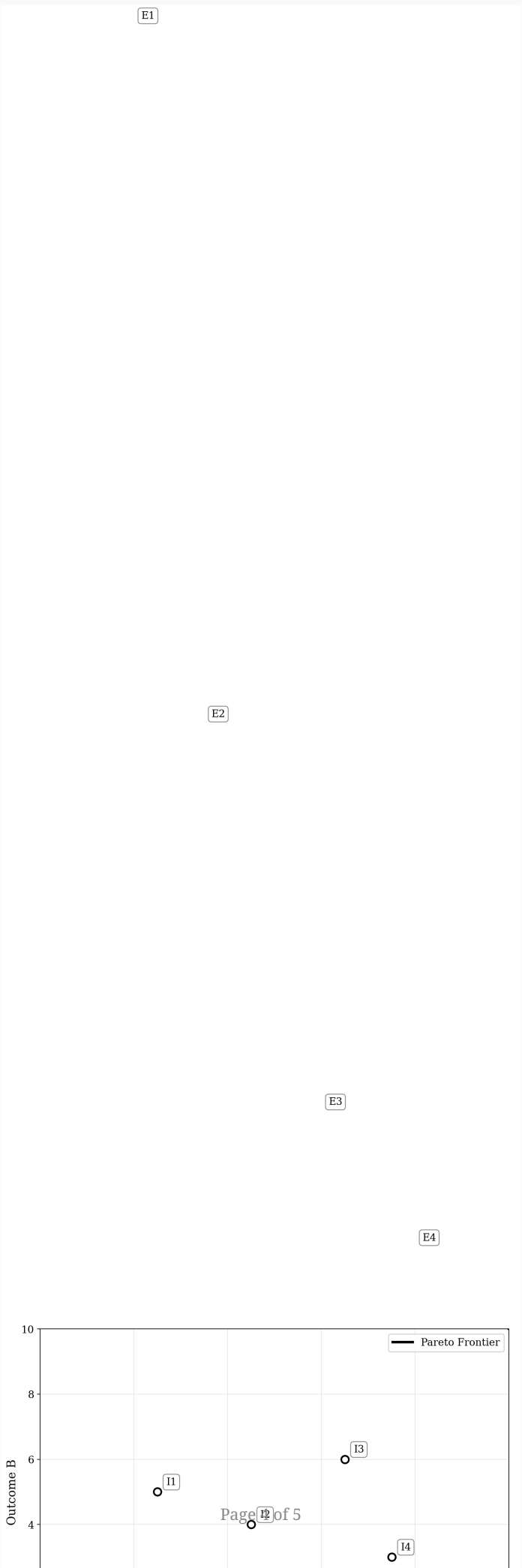
E1
E2
E3
E4
Pareto Frontier
I1
I2
I3
I4
Outcome B
10
8
6
4

Fig. 5. CUMULATIVE LEAK CORRECTIONS (all must be fixed): The diagram labels the points as "E" for efficient; The diagram labels points with "E" for efficient and. A Pareto frontier diagram illustrating efficient vs inefficient outcomes. The diagram shows a curve representing the Pareto frontier, with points on the curve labeled as Pareto efficient and points below the curve labeled as inefficient. Arrows indicate potential shifts in the curve to illustrate changes in efficiency. IMPORTANT correction needed: The diagram labels the points as "E" for efficient. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT correction needed: The diagram labels points with "E" for efficient and. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

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